

Instructions

1. Use `hr.table_name` prefix for all table references (e.g., `hr.employees`)
2. Oracle SQL syntax is required - this exam uses Oracle Database via FreeSQL
3. You may use Oracle traditional syntax (comma joins, equijoins, (+) outer join) OR ANSI JOIN syntax - both are acceptable
4. All queries must run without error in FreeSQL to receive full marks
5. Use meaningful column aliases for all computed columns
6. Include ORDER BY where specified in the question
7. COMMIT all DML operations unless stated otherwise
8. Write your final answers clearly - only the last version of each query will be graded

SUBMISSION INSTRUCTIONS

Step 1 - Write Your Answer in FreeSQL Code Editor, For each question, write your SQL query in the FreeSQL SQL Worksheet.

IMPORTANT: Every answer MUST start with a comment line indicating the question number.

```
-- Q1  
SELECT ...  
FROM ...
```

```
-- Q2  
SELECT ...  
FROM ...
```

Step 2 - Download Your Files (2 Files Per Question = 20 Files Total) For each question, you must download TWO files:

File A - SQL Code (Editor Content)

1. After writing and running your query in FreeSQL
2. Click the Download button in the toolbar
3. Select Download Editor Content
4. This saves your `.sql` file

File B - Query Results

1. After running your query successfully
2. In the Query Result tab at the bottom panel
3. Click Download then select JSON
4. This saves your result set

Step 3 - Rename Your Files

Rename each file using this exact naming format:

Format: `Nama_NomorNIM_NomorSoal`

Example:

```
JohnDoe_2501234567_Q1.sql  <- SQL code for Question 1  
JohnDoe_2501234567_Q1.json <- Results for Question 1  
JohnDoe_2501234567_Q2.sql  <- SQL code for Question 2  
JohnDoe_2501234567_Q2.json <- Results for Question 2
```

Step 4 - Upload to Google Drive or OneDrive

1. Create a new folder in your Google Drive or OneDrive
2. Name the folder in this format: NIM_NAME

Step 5 - Set Folder Access to Open to All

Google Drive:

1. Right-click the folder → Share
2. Under General Access → change to Anyone with the link
3. Set permission to Viewer
4. Click Copy link

OneDrive:

1. Right-click the folder → Share
2. Select Anyone with the link can view
3. Click Copy

TEST YOUR LINK: Open the link in an incognito/private browser window BEFORE submitting. If you can see the files without logging in, it is correct.

Step 6 - Submit the Folder Link to Binus Maya

1. Go to Binus Maya → Your course page → Assignment
2. Paste your folder link as your submission answer
3. Click Submit before the deadline

If the lecturer cannot access your link, your submission will be considered incomplete.

<5 PTS> Question 1 : You are presenting a payroll analysis to the Finance Director. She needs a clean, filtered summary to identify departments with significant salary disparity.

Task : Write a SQL query that produces a department payroll summary report with the following requirements:

1. Show each department name (not ID)
2. Show the headcount (number of employees)
3. Show the average salary rounded to the nearest whole number
4. Show the total monthly payroll (sum of salaries)
5. Show the salary spread ratio-calculated as: MAX salary ÷ MIN salary, rounded to 2 decimal places
6. Filter: Only include departments with:
 - o More than 3 employees, AND
 - o A salary spread ratio greater than 1.5
7. Order by total payroll descending

Note: Handle the case where MIN salary could be zero to avoid division errors.

<5 PTS> Question 2 : The COO needs a global workforce report that includes every employee's location, role, salary, and whether their pay is within the approved job band. This will be used for an HR compliance audit.

Task : Write a SQL query that produces a full employee compliance report:

1. Show employee full name (First Last format)
2. Show job title
3. Show department name-show 'Unassigned' if no department
4. Show city-show 'N/A' if no location
5. Show employee salary
6. Show a Compliance Status column:
 - o 'BELOW BAND' if salary < job's min_salary
 - o 'ABOVE BAND' if salary > job's max_salary
 - o 'WITHIN BAND' otherwise
7. Include ALL employees-even those without a department
8. Exclude employees from the 'Shipping' department
9. Order by compliance status, then employee last name

Note: Must use at least 4 tables and handle NULL departments correctly.

<15 PTS> Question 3 : The Compensation Committee has approved two salary adjustments that must be implemented immediately in the HR system.

Task : Write a complete DML transaction that performs the following:

Update 1:

- All employees in IT department (dept_id = 60) whose salary is below the IT department's average salary receive a 10% salary increase
- Round the new salary to the nearest whole number

Update 2:

- All employees in Sales department (dept_id = 80) who currently have no commission (commission_pct IS NULL) are assigned a commission rate of 0.10

After both updates:

- Write a verification SELECT that shows employee_id, first_name, salary, and commission_pct for all employees in departments 60 and 80, ordered by department_id then salary
- COMMIT the transaction

Note: The average for Update 1 must be calculated dynamically using a subquery-do not hardcode the number.

<10> Question 4 : HR Leadership wants a **career timeline** for succession planning, a chronological view of every role each high-potential employee has held, combining current and historical data.

Task : Using **UNION ALL**, create a **unified career timeline** that combines current and historical positions:

Include only employees who have at least one job_history record.

For each position (current or historical), show:

1. **Employee full name** (First Last)
2. **Job title**
3. **Department name**
4. **Start date** : formatted as 'DD-MON-YYYY'
5. **End date** : formatted as 'DD-MON-YYYY', or the word 'PRESENT' for current roles
6. **Record type** : 'CURRENT' or 'HISTORICAL'

Order by employee name, then start date ascending.

*Note : Both SELECT statements in UNION ALL must have the **same number of columns** with compatible data types.*

<15 PTS> **Question 5** : The HR Director wants a **reusable management dashboard** that she can query anytime without having to re-write a complex query. She specifically wants to see which managers are managing large teams.

Task : Complete the following two steps:

Step 1 Create a View:

Create a view named **manager_dashboard_v** that shows **only employees who are managers** (i.e., at least one other employee reports to them). The view must include:

- employee_id
- manager's full name (First Last)
- job title
- department name
- salary
- **number of direct reports** (employees whose manager_id = this employee's employee_id)

*Note : Use **EXISTS** to identify who is a manager, and a correlated subquery in SELECT for the count of direct reports.*

Step 2 Query the View:

After creating the view, write a SELECT that:

- Returns managers with **more than 3 direct reports**
- Shows all columns from the view
- Orders by direct_reports descending

Scoring :

5 pts	Query produces correct and complete results
3 pts	Professional output: proper aliases, formatting, ORDER BY
2 pts	Edge cases handled correctly (NULLs, empty groups, divide by zero)

*Please review all your answers before submitting.
Good luck!*